Chapter-3

Relational Data Model

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Contents

- Relational Model Concepts
- Relational Constraints & Relational Database Schema
- Relational Operations
- Relational Languages and Views
- Relational DBMS

Relational Model Concepts

- The relational model for database management is an **approach to** logically **represent** and **manage the data** stored **in a database**.
- A **database implemented and organized** in **terms of the relational model** is known as a relational database management system (**RDBMS**).
- In this model, the data is organized into a **collection of two-dimensional inter-related tables**, also known as **relations**.
- Each relation is a collection of **columns** and **rows**, where the **column** represents the **attributes** of an entity and the **rows** (or **tuples**) represents the **records**.

Terminologies

- **Relation:** Two-dimensional table used to store a collection of data elements.
- **Tuple:** Row of the relation, depicting a real-world entity.
- **Attribute/Field:** Column of the relation, depicting properties that define the relation.
- **Domain:** Set of **pre-defined atomic values** that an attribute can take i.e., it describes the legal values that an attribute can take.
  - **Eg.:** **Name** : string of characters that represent name of persons.

    Employee- age: possible ages of a company (20 – 70 years old)
  - **It also specifies the data type e.g. age must be positive integers**

Cont…

- **Relational Schema:**
  - It is the **logical blueprint** of the **relation** i.e., it describes the design and the structure of the relation.
  - **Made up of a relation name R and a list of attributes A1, A2, .., An**
  - It contains the **table name**, **its attributes**, and their **data types** (optional):
    - TABLE_NAME(ATTRIBUTE_1:TYPE_1,ATTRIBUTE_2TYPE_2, ...)
  - **For Student relation example, the relational schema will be:**
    - STUDENT(ROLL_NUMBER: INTEGER, NAME: string(20), CGPA: FLOAT)

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno

- **Degree/ arity:** It is the **total number of attributes** present in the relation schema. For the above example the **degree of the relation is 3.**

Cont…

- **Cardinality:** It specifies the **number of entities** involved in the relation i.e., it is the total **number of rows** present in the relation.
- **Relational Instance/ relation state:** It is the **collection of records** present in the relation **at a given time** (set of tuples at a given time) .
- **Relation Key:** It is an **attribute** or a group of **attributes** that can be **used to uniquely identify** an entity in a table or to determine the relationship between two tables. Relation keys can be **of 6 different** types:
  - Candidate Key, Super Key, Composite Key, Primary Key, Alternate Key, surrogate key, Unique Key and Foreign Key

Cont…

Primary key

- A primary is a single column value used to identify a database record uniquely.
- It has following attributes
- A primary key cannot be NULL
- A primary key value must be unique
- The primary key values should rarely be changed
- The primary key must be given a value when a new record is inserted.

Composite Key

- A composite key is a primary key composed of **multiple columns used to identify a record uniquely**

Note:

1. A Relation is a collection of rows (tuples) and columns (attributes).
2. In a relation, the **tuples depicts real-world entity**, while the **attributes** are the **properties** that **define the relation**.
3. Structure of the relation is described by the **relational schema.**
4. **Relational keys** are used to **uniquely identify a row** in a table or to **determine the relationship between two tables**.

Cont…

- ✓ **Example**: Domains, Attributes, Tuples, and Relations (all in one)

Relation Name

Attributes

STUDENT

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.53
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	3.25
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.93
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	2.89
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.21

Tuples

Characteristics of Relations

- **Ordering of Tuples within Relation is not important**
  - A relation is defined as a set of tuples.
  - Mathematically, elements of a set have no order among them, hence, tuples in a relation do not need to have any particular order.
  - Duplication is not allowed, that is why we use keys while constructing relations.
  - In other words, a relation is not sensitive to the ordering of tuples.
- **Ordering of values within a tuple is important**

Cont…

- Values & NULLs in the Tuples
  - Each **value in a tuple** is **an atomic** value so, **composite (e.g. Address)** and **multivalued attributes** are **not allowed.**
  - This model is sometimes called the **flat relational model.**
  - Much of the **theory behind the relational model** was **developed with this assumption in mind,** which is called the **first normal form assumption**.
  - Hence, **multivalued attributes** must be **represented** by **separate relations,** and **composite** attributes are **represented** only by their **simple component attributes** in the basic relational model.
  - An important concept is that of **NULL values** which **represents** the values of attributes that may be **unknown** or **may not apply** to a tuple.
  - A special value called NULL is used in these cases.

Cont…

- **Interpretation of a Relation**
  - **The relation schema can be interpreted as a declaration or a type of assertion.**
    - E.g; The schema of the STUDENT relation of (slide 8) asserts that, in general, a student entity has a Name, Ssn, Home-phone, Address, Office-phone, Age, and Gpa.
  - **Each tuple in the relation can then be interpreted as a fact or a particular instance of the assertion.**
    - E.g; The first tuple in (slide 8) asserts the fact that there is a STUDENT whose name is Benjamin, Ssn is 305-61-2435, Age is 19 and so on...

Cont…

- Notice that some relations **may represent facts** about **entities**, whereas other **relations** may **represent facts** about different **relationships**.
- E.g; A relation schema MAJORS (student-ssn, department-code) asserts that students major in academic disciplines (student & department relations).
- A **tuple** in this relation **relates** a **student** to **his or her major discipline/field of study.**

Relational Model Notation

- We can use the following sample notations for our operations:
  - A relation schema R of degree n is denoted by R(A1, A2,.., An).
  - The name of a relation schema such as STUDENT indicates the current set of tuples in that (the student) relation.
  - An attribute A can be qualified with the relation name R to which it belongs by using the dot notation R.A.
    - **For example:** STUDENT.Name, TEACHER.Age.
  - This is because the same name may be used for two attributes in different relations.

Cont…

- However, **all attribute names in a particular relation must be distinct.**
- An **n-tuple t** in a **relation r(R)** is denoted by **t = [v1, v2,... vn],** where **vi** is the **value corresponding** to **attribute Ai**.
  - E,g; Consider tuple t = [’Barbara Benson’,‘533-69-1238’, ‘(817)839-8461’, ‘7384 Fontana Lane’, NULL, 19, 3.21]
  - From the STUDENT relation in the example; we have: t[Name] = [‘Barbara Benson’], and t[Ssn, Gpa, Age] = [‘533-69-1238’,3.21, 19].

Relational Model Constraints & Relational Database Schemas

- In a **relational database**, **there will** typically be **many relations**, and **the tuples in those relations are usually related in various ways**.
- The state of the whole database will correspond to the states of all its relations at a particular point in time.
- **There are generally many restrictions or constraints on the actual values in a database state.**
- Constraints are a **restriction or limitation** of data in the database

Con't…

- Constraints on databases generally have three main categories:

1. **Inherent model-based constraints or implicit constraints:**

- Constraints that are inherent in the data model

  Example → the constraint that a **relation cannot have duplicate tuples** is an inherent constraint.

Con’t…

2. Application-based /semantic constraints or business

- Constraints that **cannot** be directly **expressed** in **the schemas of the data model**, and hence must be **expressed and enforced by the application programs** some other way. How good the relational database is?

Cont…

3. Schema-based constraints or explicit constraints: Constraints that can be directly expressed in the schemas of the data model, typically by specifying them in the DDL.

Eg.: Age of an employee has to be between 20 -45 years so age is restricted. It directly related to relational schema

Cont…

- Schema-based constraints

    1. Domain constraints,
    2. Key constraints and constraints on NULLs,
    3. Entity integrity constraints, and referential integrity constraints

Cont…

- **Domain Constraints:**
  - Domain constraints specify that within each tuple, the value of each attribute has to be an atomic value/ indivisible and must have single attribute name.
  - This constraint checks the datatype of the values of each attribute

STUDENT	RollNo	Name	Phone	Age
	1	Jeremy	2563251425	14
	2	Charles	1234568978	14
	3	Weston	3625368914	A

Violates Domain Constraints

Key Constraints and Constraints on NULL Values

Key constraints

- An attribute that can uniquely identify each tuple in a relation is called a **key**.
- A super key specifies that **no two tuples** can have the **same value.**
  - It is any set of attributes that helps to identify each tuple uniquely.
  - Every relation has at least one super key → set of all attribute.

Student table

RollNo	Name	Age	Email
1	Yeabsira	18	yabita@gmil.com
2	Samuel	18	sami12@gmail.com
3	Haleta	20	Haleta05@gmail.com

SK = {RollNO}, {Email}, {RollNO, Name}, {RollNO,Age}
{RollNO, Email}, {Name, Email}, {Age, Email},
{RollNo, Name, Age, Email}

Cont…

- A key must satisfy two properties:

1. Two distinct **tuples in any state** of the relation **cannot have identical values** for **all the attributes** in the relation. → This uniqueness property **applies for both key and a superkey.**
2. It is a **minimal superkey (candidate key)** that is, a superkey from which we cannot remove any attributes and still have the uniqueness constraint hold.
   - This **minimality property** is required for **a key** but is **optional for a superkey.**
   - For e.g. in previous relation RollNo or email can be a key
   - Hence, a **key is a superkey but not vice versa**

Cont…

- A **relational schema can have more than one key**, each of the keys is called **candidate key**
- It is a **set of attributes** that **uniquely** identify the **tuples** in a relation.
- We can have any number of candidate keys in a relation but **one is taken as primary key**
- The key **whose values are used to identify tuples** in the relation is a primary key (PK).
- It is usually **selected by database designers.**
- **Eg: RollNo and Email are candidat keys** for student relation

Reading Assignment

- What is candidate key, composite key, alternate key & surrogate key, and their advantages in real world?
- What is the difference between candidate and super key?

Key Constraints and Constraints on NULL Values cont…

Constraints on NULL value

- Specifies whether null values are permitted or not (NOT NULL)

Entity integrity constraint & Referential integrity constraint

Entity integrity constraint

- Specifies that **no primary key value can be null**
- Having NULL values for PK → we cannot identify some tuples.
- For example: If two or more tuples had NULL for their PKs, we may not be able to distinguish them if we try to reference them from other relations.
- **Entity integrity constraints** and **NOT NULL** are **specified** on **individual relations or tables.**

Cont..

Referential integrity constraint

- Specified **between 2 relations and** is **used to maintain the consistency among tuples** in the two relations.
- States that a **tuple in one relation** that **refers** to **another relation** must **refer** to **an existing tuple** in that relation.

Referential integrity constraint Cont…

STUDENT

SID	SNAME	DNO
1001	Yeabsira	2
1002	Samuel	1

The attribute of one relation that refers to the primary key of another relation is known as **foreign key**

DEPARTMENT

DNO	DNAME
1	CS
2	SE
3	IS

Referential Integrity constraint & Foreign Keys

- An attribute FK in a relation schema R1 is a FK of R1 that references relation R2 if it satisfies the following rules:
  1. The attribute in FK of R1 must have the same domain(s) as the primary key attributes PK of R2 → the attributes FK are said to reference or refer to the relation R2 or in other word It has to be the same domain as the attribute it refers to

Cont ..

2. A value of FK in a tuple t1 of the current state r1(R1) either occurs as a value of PK for some tuple t2 in the current state r2(R2) or NULL.

- i.e. we have t1[FK] = t2[PK], and we say that the tuple t1 references or refers to the tuple t2. **(t represents tuple)**
- In this definition, R1 is called the referencing relation and R2 is the referenced relation i.e. for previous example **student relation** is **referencing relation** where as **department relation** is **referenced relation.**

- If these two conditions hold, a referential integrity constraint from R1 to R2 is said to hold.

Cont….

- In general, a **relation schema may have more than one unique identifier attribute (key).**
  - **In this case, each of the keys is called a candidate key.**
- E.g: A CAR relation (on next slide) has **two candidate** keys: **License-number** and **Engine-serial-number.**
- It is **common to designate one of the candidate keys as the PK** of a relation.

- The key whose values are used to identify tuples in the relation is a primary key (PK).
- It is usually selected by database designers.
- We use the convention that the **attributes that form the primary key** of a relation schema are **underlined.**

Cont….

CAR

License_number	Engine_serial_number	Make	Model	Year
Texas ABC-739	A69352	Ford	Mustang	02
Florida TVP-347	B43696	Oldsmobile	Cutlass	05
New York MPO-22	X83554	Oldsmobile	Delta	01
California 432-TFY	C43742	Mercedes	190-D	99
California RSK-629	Y82935	Toyota	Camry	04
Texas RSK-629	U028365	Jaguar	XJS	04

fig: 2.2

Relational Databases & Relational Database Schemas

- A relational database usually contains many relations, with tuples in relations that are related in various ways.
- A relational database schema S is a set of relation schemas S = {R1, R2, . . . , Rm } and a set of integrity constraints IC.
- A relational database state DB of S is a set of relation states DB={r1, r2,...} such that each ri is a state of Ri and such that the ri relation states satisfy the integrity constraints specified in IC.
- Fig 2.3 shows a relational database schema that we call COMPANY = {EMPLOYEE, DEPARTMENT, DEPT-LOCATIONS, PROJECT, WORKS-ON, DEPENDENT}.

Cont…

- Each relational DBMS must have a data definition language (DDL) for defining a relational database schema.
- Current relational DBMSs mostly use SQL for this purpose.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date

DEPT_LOCATIONS

Dnumber	Dlocation

PROJECT

Pname	Pnumber	Plocation	Dnum

WORKS_ON

Essn	Pno	Hours

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship

Fig 2.3

Entity Integrity, Referential Integrity & Foreign Keys

- E.g; In fig 2.4, the attribute **Dno of EMPLOYEE gives** the **department numbe**r for which **each employee** works;
  - Hence, its value in every EMPLOYEE tuple must match the Dnumber value of some tuple in the DEPARTMENT relation.

Entity Integrity, Referential Integrity & Foreign Keys

- Referential integrity constraint diagrammatic representation

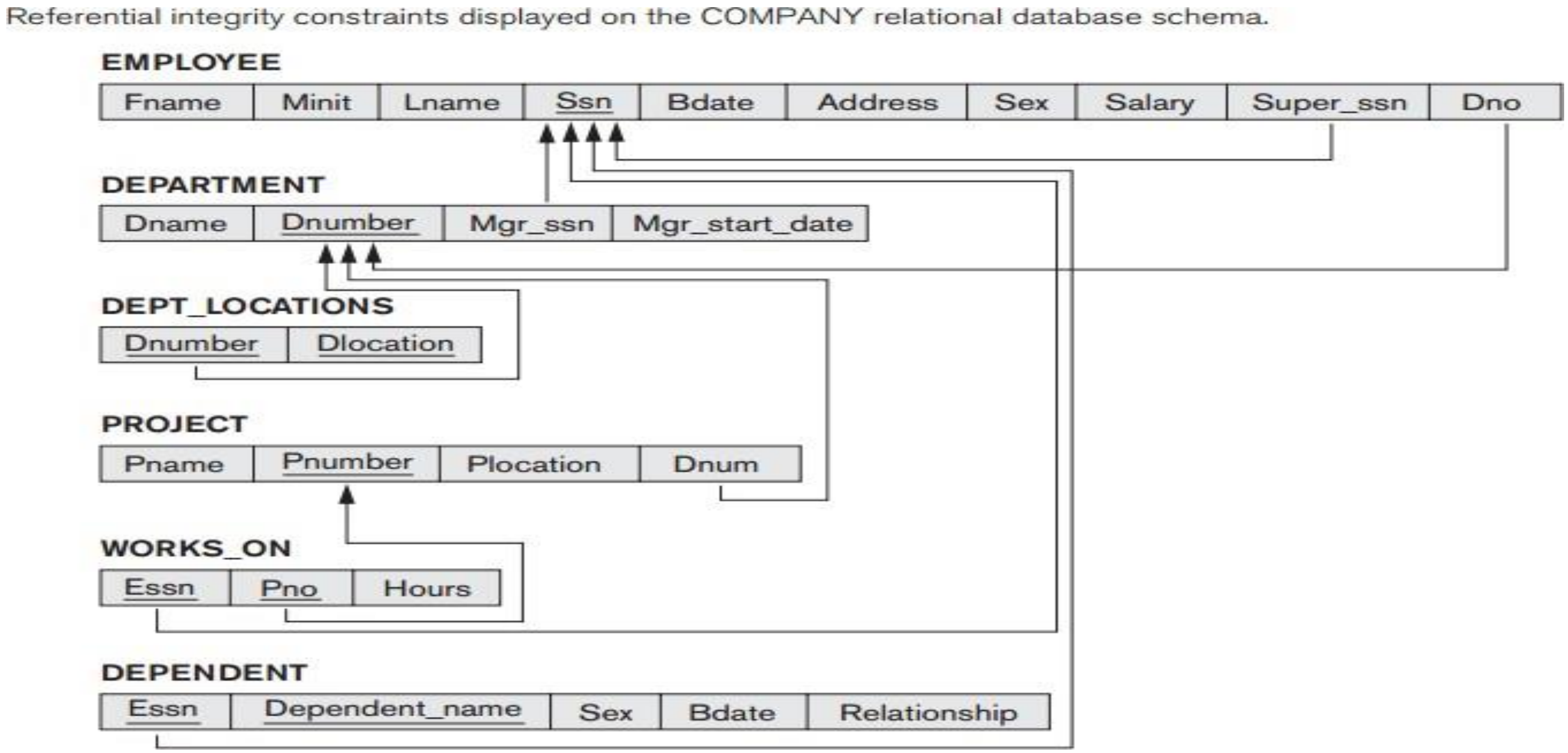
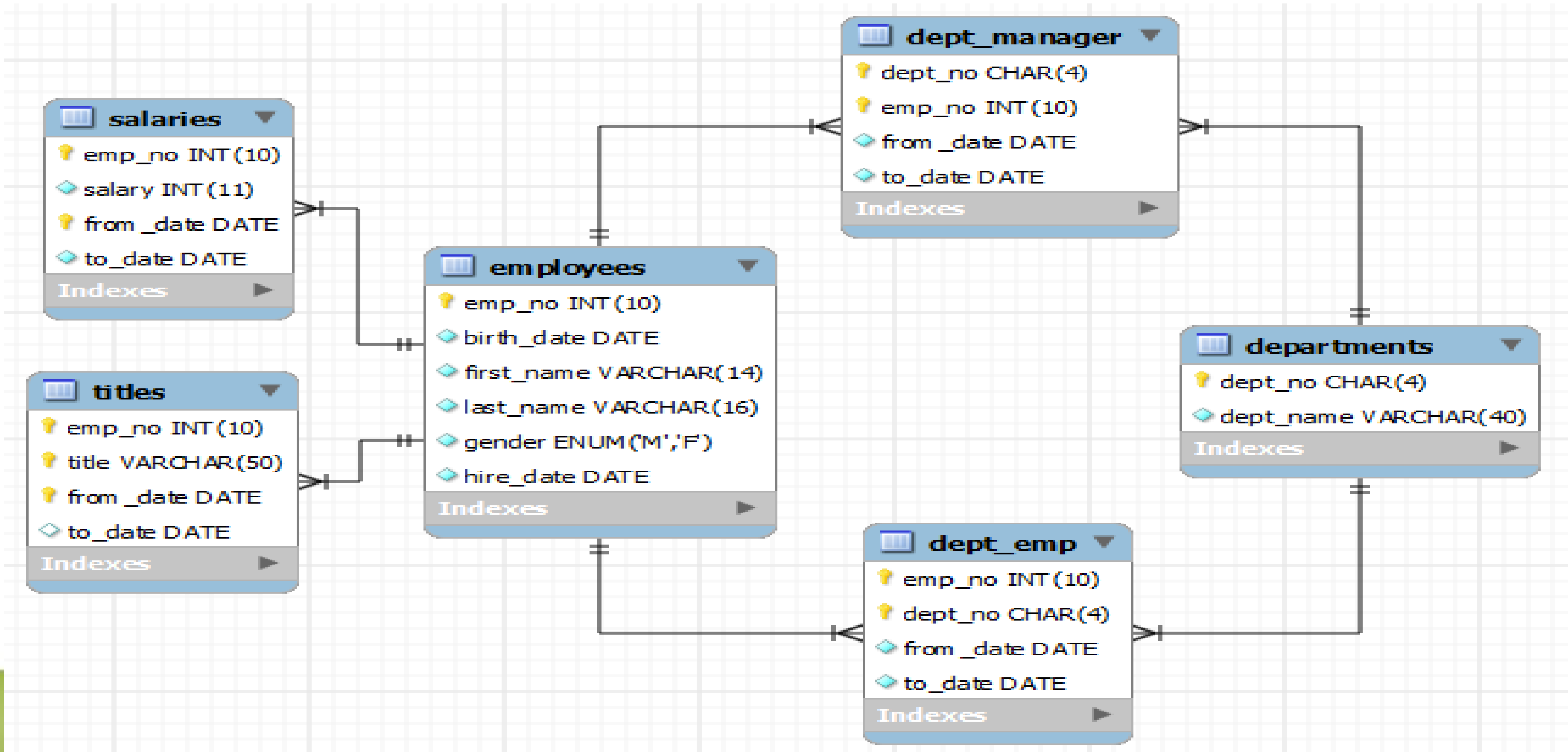


Fig 2.4

Database Design Models and Classification: RDBMS

A database model: is a model that determines the logical structure of a database and fundamentally determines in which manner data can be stored, organized and manipulated.

E.g;



Reading assignment

What are Composite Key and Alternate Key?

Which types of DBMSs are the Best In 2022 and 2023? Why?

1. Draw ER diagram for the following scenario

There is a company called SMS bakery. Some employees in this company has a responsibility to bake a cake for the company. The company is managed by single manager. Each employee has name, Id, birthdate, age, certificate, address, phone number, and working roll. Cake has type, size, order number and price. The company has number of customers and One customer may order more than one cake at a time. Customer has phone no, address, name.

A. Identify and present each entity types clearly

B. Identify and present each attribute (consider attribute types like primary key)

C. Identify and link all the relationships (remember to consider relationship constraints i.e cardinality ratio and participation constraints)

2. Show the relational schema for each entity type

Thank you !!!